Deep Learning Based Visual Feature Extraction from Document Images Using VGG16

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***Abstract*—Document image understanding is an important research area in document analysis and information retrieval. This work presents a deep learning framework based on the VGG16 convolutional neural network for visual feature extraction from document images. The model extracts hierarchical visual representations including edges, corners, logos, tables, text regions, and document layouts. Transfer learning is utilized to improve classification performance and reduce training time. Experimental evaluation demonstrates the effectiveness of the proposed approach for document image classification.**

***Index Terms*—Document Analysis, Visual Feature Extraction, Deep Learning, VGG16, CNN, Transfer Learning.**

I. Introduction

Document image classification is widely used in digital archiving, automated document processing, and information retrieval systems. Traditional approaches rely on handcrafted features that often fail to capture complex document structures. Deep learning techniques automatically learn hierarchical visual representations directly from document images and have significantly improved classification performance.

This work employs a VGG16-based transfer learning framework to extract meaningful visual features from document images.

II. Problem Statement

The objective is to develop a deep learning model capable of extracting visual features from document images. The extracted features should capture:

- Edges and corners
- Logos and photographs
- Text regions
- Tables and forms
- Structural document layouts
- Texture information

The extracted features are subsequently used for document classification.

III. Dataset Description

The dataset consists of document images organized into category-specific folders. Images are loaded using the PyTorch ImageFolder utility.

A. Dataset Reference

The Tobacco800 dataset has been used as a benchmark dataset for document image analysis.

Dataset Link:

https://www.kaggle.com/datasets/sprytte/tobacco-800-dataset

IV. Data Preprocessing

The preprocessing steps include:

1) Resizing images to 224×224 pixels.
2) Conversion of images to tensors.
3) Splitting the dataset into training, validation, and testing sets.

V. Proposed Methodology

The proposed framework consists of the following stages:

1) Dataset loading
2) Image preprocessing
3) Feature extraction using VGG16
4) Transfer learning
5) Model training
6) Validation and testing
7) Performance evaluation

VI. Model Architecture

The VGG16 architecture consists of:

- 13 convolutional layers
- ReLU activation functions
- 5 max pooling layers
- 3 fully connected layers

The final fully connected layer is modified according to the number of document classes.

Architecture flow:

Input Image $\rightarrow$ Convolution Layers $\rightarrow$ Pooling Layers $\rightarrow$ Feature Maps $\rightarrow$ Fully Connected Layers $\rightarrow$ Classification

VII. Implementation Details

The implementation is performed using the PyTorch deep learning framework.

A. Data Transformation

```
transform = transforms.Compose([
transforms.Resize((224,224)),
transforms.ToTensor()
])
```

B. Loss Function

```
criterion = nn.CrossEntropyLoss()
```

C. Optimizer

```
optimizer = optim.Adam(
model.parameters(),
lr=0.0001
)
```

VIII. Training Configuration

TABLE I
TRAINING PARAMETERS

Parameter	Value heightArchitecture
VGG16 heightImage Size	224 $\times$ 224 heightBatch Size
16 heightOptimizer	Adam heightLearning Rate
0.0001 heightLoss Function	Cross Entropy Loss heightEpochs
10 height	

IX. Visual Feature Extraction

The convolutional layers of VGG16 are utilized as a feature extractor.

```
feature_extractor = model.features

with torch.no_grad():
features = feature_extractor(images)

print(features.shape)
```

The extracted feature tensor typically has dimensions:

[FeatureMap $\in R^{512\times7\times7}$]

These feature maps encode high-level visual information including text blocks, logos, tables, and structural document patterns.

X. Alternative Approaches

Alternative deep learning architectures include:

- ResNet50
- DenseNet121
- EfficientNet
- Vision Transformer (ViT)
- Dilated Convolution Networks

Dilated convolutions increase the receptive field and can capture larger document structures more effectively.

XI. Experimental Results

The model is evaluated using standard classification metrics.

A. Performance Metrics

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

XII. Discussion

The VGG16 model successfully learns discriminative visual representations from document images. Transfer learning improves convergence speed and reduces computational cost compared with training from scratch. The extracted feature maps capture both local visual patterns and global document structures.

XIII. Conclusion

This work presented a VGG16-based visual feature extraction framework for document image analysis. Experimental results demonstrate the capability of deep convolutional neural networks to extract meaningful document representations and perform accurate document classification.

Future work may investigate dilated convolutions, attention mechanisms, and transformer-based architectures to further improve document understanding.

References

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